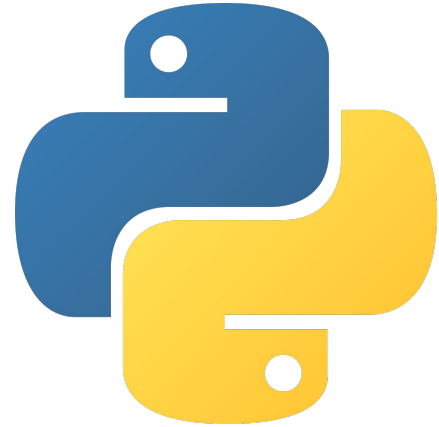




Python Week 3

Loops & Repetition





Objective: Making your code run again...and again...

What You'll Learn:

- How to use loops to repeat code
- Difference between for loops and while loops
- Combine loops with if statements to control what gets printed



Do Now (Week 2 Review)

Write a program that:

1. Asks the user for a number.
2. Prints "even" if the number is even, and "odd" if the number is odd.



```
number = int(input("Enter a number: "))

if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```



I. Introduction to Loops



Why do we use loops?

Sometimes, we want to print the same thing multiple times. It can be annoying to write:

```
print("Hello")  
print("Hello")  
print("Hello")
```



Instead, we can use loops!

- A loop repeats code multiple times.
- Saves time and effort
- Real life example: brushing your teeth back and forth until they're clean.



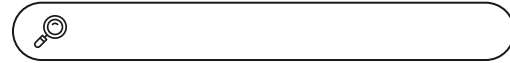
Multiple Choice: Which of the choices is NOT an example of a loop?

- A. A runner completing 4 laps around a track.
- B. A teacher taking attendance by calling each student's name one by one.
- C. A student turning on a light switch once when they enter a room.
- D. An alarm clock that snoozes every 9 minutes until you turn it off.



Multiple Choice: Which of the choices is NOT an example of a loop?

- A. A runner completing 4 laps around a track.
- B. A teacher taking attendance by calling each student's name one by one.
- C. A student turning on a light switch once when they enter a room.
- D. An alarm clock that snoozes every 9 minutes until you turn it off.



II. For Loops



For Loop Structure

```
for i in range(3):  
    print("Hello")
```

for => starts loop

i => counter (starts from 0)

range(3) => repeats 3 times



This loop runs three times, printing "Hello" three times:

0 => Hello

1 => Hello

2 => Hello

```
for i in range(3):  
    print("Hello")
```



Multiple Choice: What does the following code print out?

```
for i in range(3):  
    print(i)
```

- A. 1 2 3
- B. 0 1 2
- C. 0 1 2 3
- D. Error



Multiple Choice: What does the following code print out?

```
for i in range(3):  
    print(i)
```

- A. 1 2 3
- B. 0 1 2**
- C. 0 1 2 3
- D. Error



Range with start/end

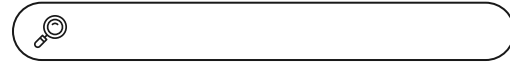
```
for i in range(1, 6):  
    print(i)
```

Outputs: 1 2 3 4 5



Try it: Write a code that prints out your name three times.

```
for i in range(3):  
    print("Hello")
```

III. While Loops



While Loop Structure

```
count = 1

while count <= 3:
    print(count)
    count += 1
```

What is the count after
the first loop?

Start: 1

Print

Add 1 => now 2



While Loop Structure

```
count = 1  
  
while count <= 3:  
    print(count)  
    count += 1
```

“Repeat while
something is TRUE”



Be careful of infinite loops!

```
while True:  
    print("Hi")
```

Q; What's wrong with this code?



IV. Loops + If Statements



Combining loops + If

```
for i in range(1, 6):  
    if i % 2 == 0:  
        print(i)
```

i = 1 → skip

i = 2 → print

i = 3 → skip

i = 4 → print



Try it: Write code that prints numbers 1-10, but only numbers greater than 5.



Try it: Write code that prints numbers 1-10, but only numbers greater than 5.

```
for i in range(1, 11):  
    if i > 5:  
        print(i)
```




V. Nested Loops



A nested loop is a loop inside another loop.

Q: How many times does this run? A. 2, B. 3, C. 4

```
for i in range(2):  
    for j in range(2):  
        print(i, j)
```



Now it's your turn!

Write a Program that:

1. Asks the user for a number.
2. Uses a for loop to print numbers from 1 up to that number.





Classwork Answer:

```
num = int(input("Enter a number: "))  
  
for i in range(1, num + 1):  
    print(i)
```



Thanks!

Have questions? Reach out:

cs4nyc0@gmail.com

cs4nyc0.wixsite.com/code4nyc



@code4nyc_



@code4nyc